

Active InferAnt Stream 005.1 ~ FORMINDEX: FORMIS

INtegrated Database EXploration

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all right hello and welcome everyone this is active infant stream number 5.1 October 16th 2024 we're going to begin with a GitHub push I'm going to check it's been updated in GitHub this is also published on zenodo so it has a DOI it's citable and to show how to get a DOI on zenodo I'm going to make release 0.3 right now adding a new tag 0.3 infer stream 5.1 publish it published on GitHub come back to zenodo and we can find it citable it just takes a few minutes for it to Archive it all right so here we are I have some topics and sections to go through and then if you're watching live please feel free to write any comments and questions in in the live chat and this should should be pretty interesting there's going to be a lot of ant bibliographic methods as well as some generative AI synthetic intelligence methods usage of perplexity llms and so on okay so we began with a welcome push the live stream update to GitHub we'll give a brief overview of the form index project I'm using cursor version 0.423 made the new version and the repo is list Ed here so what is the form index project here we are in the form index Coda which is linked in the video description so the form index is a multi-stage project related to ant literature so in ant literature there is the formis or formi depending on how you pronounce it and this was a effort is an effort that curated citations related to ants and it has uh completeness in taxonomic areas of ant studies so a lot of rare ant taxonomy literature complete up to 1996 and then not as much after and so in for andex I'm setting out on on a dual journey to do an analysis of formis for its own internal sense making to understand formis as a literature resource itself and then also to look more broadly towards integration of knowledge methods for micology for ant research and then eventually even go beyond that with meta informat and integrating bioinformatics and so on so in this first part of the stream it's going to begin with the formis analysis which is going to briefly move through the ant specific Parts about how the bib text bibliography with 80 th000 references is brought in and then some of the visualizations that come out of the just looking at formis itself then the second part of the methods will focus on generative Ai and we'll get there when we get there

so first for the formis analysis so first Sanford sent 316 sample citations and on October 8 I read in those just to ensure I could get the format going he then sent over the bib text plain text file with 880,000 records which I brought in in the script section with these scripts read in formis and visualize we'll run them so first reading in forus we run this read in foris this was from another computer so we'll see let's fix it on the go meanwhile though continuing on the report this script reads in the bib text and stores the Json format the generate Target bibliographies subsets the 80,000 row Json database into uh bibliographies that mention that term so this is what it looks like for bioenergetics it's a Jason that's a strict subset of the overall we'll focus more on running the code for the second part that I've more recently done on this computer the visualize form. py see if this one works also this one doesn't work but for the all formis database and for every Target bibliography some kinds of images are made so here's the distribution of Records here's histogram of record types so there's 60,000 articles 11,000 Tech reports 5,000 collections 3,500 PhD thesis 1592 books one miscellaneous here's authors by number of citations an non coming through jhc deer only slightly behind Publications per year so as mentioned earlier this is not the overall number of Publications per year in Google Scholar or literature database this is just for formis and this reflects what could be a really interesting pattern to continue exploring as we integrate other literature searches we know that ant Publications did continue after 1996 and there's other citations so it's an interesting question how do we add in what should be the scope of this database what what keeps it ant relevance what's the relevance realization for the ant literature here's a co-authorship heat map some of these could be cleaned up with uh different visual improvements it's showing here though the number of co-authorships between different authors so pretty cool to identify all these diads and in the literature we could even look for higher order authorship combos here's top venues so locations for publishing insect soci show evolution of the venues Through Time locations wordcloud for the title hotra for missier and in the abstract wordcloud so those are the visualizations that's just to get pretty fast through the formis method and get to the generative AI part where we're going to have active inference and all these other fun applications of of AI so there's going to be three AI generative AI method sections here notebook LM open AI API and then perplexity and there's going to be the most to explore in perplexity so first notebook LM here I took the target bibliography subsets and in the Json format uploaded to a new notebook at notebook LM made a made a podcast conversational style from those citations then uploaded it to Revit AI to make these so all mute my local microphone and just let's see if the audio will come through just to hear what it's like for a

second ever looked at an anthill and thought what's the story there H well today yeah we're going full Anthropologist on that I like it we're diving into the world of the red harvester ant okay or Pagan Barbados okay awesome so follow on through the links to hear the podcasts and really promising for local text to speech generation in the future okay second is the closest to what was previously done in some of the earlier infan streams where I used the open AI API to concatenate the target bibliographies with a general summarization prompt that's the the pro summary just the concatenated text file kind of like the prepr grants from early on and then script two uses the openai API to get the summary of the literature and then script three takes in the literature summaries and translates into Target languages so this is more examples of how different literature resources different different text inputs can be summarized for different audiences and in different human languages okay now we're going to get to the fun and the new stuff with perplexity a thank you Tucker for your message making you a moderator for that one I hope you enjoy your new Ranch Tucker okay so just to get familiar with what perplexity is here I pasted a prompt into perplexity doai write a comprehensive summary of the RX INF fur. JL package and perplexity goes through a series of steps it does online searches finds relevant chunks of information and then it writes this report responding to the prompt with internet information including these code boxes so I'm showing that because what this next section is going to do is essentially send this exact prompt among others and then interface with it through the scripts so here's another example create a comprehensive Grant proposal outlined for the NSF safety security and privacy of Open Source ecosystems safe o program and then again similarly it just thinks okay which which ecosystem how about tensorflow so enhancing security and privacy in the tensorflow ecosystem and then writes the grant making up the information also on the notebook LM a friend just shared with me illuminate which you can do the literature search this is illuminate. goole.com do the literature search so here I have it look up active inference and then had it pull in these four papers so it's taking one more step for you whereas with notebook LM you had to download the PDF and upload them let's now listen for the first time to illuminate I'm going to turn off my microphone and it will play from the computer this conversation is powered by Google illuminate check out illuminate. google.com for more Welcome to our discussion on active inference a fascinating framework for understanding how agents make decisions in uncertain environments we'll be exploring several research papers that delve into the intricacies of active inference comparing it to the well-established field of reinforcement learning what are the core tenets of active inference and how does it differ from reinforcement

learning active inference rooted in the free energy principle posits that agents act to minimize their surprise the uncertainty about the world unlike reinforcement learning which focuses on maximizing a reward signal active inference aims to minimize free energy a measure encompassing both prediction error and the complexity of the agent's internal model this another another one pretty cool though illuminate. gooogole okay in gen 24 was with different kinds of calls to action translated into a lot of language if people want to look in that repo okay so where were we with perplexity so uh I did several things in the report on the 13th and then in the several days since several other things have happened so I'm going to just skip over the earlier parts that were written about perplexity and we'll pop over to the documentation for the API and then look at how it's being used in the package now so where things are out right now pending the rest of this stream of course communicating with the formis stakeholders get their feedback get others feedback open science Etc then we have the analysis within for me looking at how we can do all kinds of augmented analyses within the database as it stands in July 2024 snapshot and then also go beyond for me July 2024 snapshot so adding in new literature connecting with the species ID perplexity searches Etc fabric integration kind of like was happening in the previous infer an street so okay all this is open source in the form index repo quick look at the perplexity API very simple docs page you get \$5 a month for paying \$20 a month wow and there's a really short simple example that gets you going here are the supported models they have at this time they have several of sizes of this perplexity sonar llama 3.1 base at 870 and 405 billion parameters they have and and uh they have several perplexity chat models these are chat they're not do online searches and then there's the instruct open source models they mention the rate limit 50 a minute we might hit that pricing is shown here and each one of these requests seems to be costing using the large model I think in most cases the the the the goldilock medium one each of these was maybe a tenth of a cent or a fifth of a Cent and I'm sure a cheaper one would be a fifth back cost and and so on so super worth it for for what we're getting all right here we go let's pop into the perplexity methods folder okay that was the formus analysis goals accessible rigorous applicable micology notebook LM open AI API for the translation now perplexity API so um test perplexity API is a short script that tests the network connectivity simple perplexity is where you can paste in your private key it's just simpler and keep it all in one just with their example of starting so this is if you want to study the simp EST perplexity call you can make so now here we get an llm keys. key it's in ignore so it's not sync to GitHub and it has the perplexity API key so there are uh as the simple perplexity shows you can send your message with a

system prompt and a user prompt so this is kind of the context or the character system prompt and then the specific user request so first the user prompts this is what somebody would type in they're in a Json called user prompts do Jason and they're numbered as well as given a short name and the full prompt so here prompt one short name is climate change impact and the prompt is conduct a comprehensive meta analysis of studies published in the last year on the impact of climate change on Ant species diversity and then extra information you can say output in this style or make sure you do a ton of research um and so there's several dozen the first few dozen are ants related then applications of ants what's the role of neurotransmitters brain regions then prompt 2425 we get into doing meta analyses of the career of Professor Deborah Gordon and Professor Tim Lings faay we we can add more if people have any requests for prompts please write them in the chat and I'll add it here uh doing meta analyses on different areas of ant Biometrics like relatedness within and across colonies phenotypic correlation measures then now prompt 33 3435 we're getting into specific ant groups doing a the last three years so cotus formica tholi adaa solenopsis monore laus okapa pogam mirx Barbados that's genus and species and then non Barbados pogam species then we get into the regional we got NorthEast Southwest USA Pacific Northwest agricultural ants Northern California ants Davis D Davis agricultural ants Davis Urban ant ecology Davis invasive ants ant symbolism ant wisdom ants and Contemporary Art ants indigenous wisdom stewardship then 65 takes a a bit of a Twist here we get more into prompts that are not necessarily related to act uh to to ants but some of them do as well rela rela to the active inference theoretical foundations recent developments applications to ant behavior of active emergence PhD proposals for active inference foraging simulators for active inference recent events related to active inference writing step-by-step tutorials this one's going to be fun to look at giving comprehensive overview of basing mechanics Quantum active inference writing a a summary and stepbystep on RX infer JL Julia package writing who's on first dialogues writing Grant proposals fire risk assessments and county level fire risk assessments thanks Tucker enjoy that's user prompts system prompts there are fewer but if someone suggests one just write it in the live chat these are also identified with a number and a short name and a longer description so here's a Mycologist some several ant researcher roles programmer complex system scientist local journalist enthusiastic undergrad system prompts seven and eight are questions from the innovator's catechism at the early ideation and then later pitch stage so they're going to respond with question and answer structured information which we'll look at and some other areas so then perplexity py you list the short name of the target system prompts and then the

target user prompts so we could add one if if somebody types another user prompt um just let's do Lake County fire risk just to show what it looks like say perplexity run it as Python 3 perplexity so where the outputs already exist then uh it skips it for that combination so it looks like that was already done let's see if it'll autoc complete or let's just pick a user prompt that hasn't been done let's do foli formica recent studies so systems uh uh innovator catechisms pitch the outputs existed for some of these user prompts already then it hits formica recent studies which is a new one so it's going to send that off to perplexity um while that's chugging away I'm going to go over to the GitHub and we're going to look at a few of the outputs of the the primary artifacts themselves so just the outputs of the perplexity different characters doing these different kinds of syntheses and then we will look at the semantic and linguistic methods that analyze the folders of documents the markdown output folder is nested into the different roles so let's compare the innovator's catechism with the complex system scientist okay here we have complex system scientist recent events here's iy number four this was a event I don't know if the dates were correct but that did happen in 20 23 there's some Institute quarterly round table so it's o o over learning on that or somehow it came to to learn about that one live stream here's basy mechanics overview this is written like a curriculum explainer here's the career meta analysis of Professor Deborah Gordon early career topics methodologies key findings this is a different Deborah Gordon both Deborah Gordons so it wasn't sure so it wrote about Deborah M Gordon the Mycologist and then other Deborah Gordon as well here's Lake County California warn your fire risk current fire conditions historical fire data climate weather pattern vegetation fuel load topography human factors infrastructural vulnerability fire management resources prevention mitigation strategies future projections economic impact ecological consideration Community preparedness postfire recovery actionable recommendations no citations provided here's overview of RX and fur this is the perplexity example on the web interface that I showed earlier here it is just in GitHub and we can test running this but this is pretty much what it looks like and if it's gener if it's pulling from the coin toss example this should be good here's the grants proposal for safe osse again that was a one where I showed the real web interface and then let's see the who's on first so this one makes a who's on on first about hurricane Milton I guess that's what it thought current events were now if we look over in the innovator's catechism system prompt folder there should be some differences here here the output format is requested to be the project catechism so with interesting madeup team names which we'll see I think a few of we see the catechism format of situation problem and this the entrepreneurial pitch catechism specifically situation in terms

of key problems user segments Alternatives early adopters the mission and the value proposition potential Avenues of approach and the resources that they're uh associated with advantages risks Fe abilities channels Milestones metrics cost benefit big picture admin folder bibliography here it's a pitch about a project to study active inference ant Behavior and the first citation is a real paper here we have innovator's catechism for the foraging simulation so and it includes some code methods here's a pitch for the basian mechanics SO pitching why we would want to have this basian mechanics explainer Border looking at these citations open up all four of them we have relevant paper with a incorrect citation ramstead at all rather than stiva devel at all second paper ramstead at all 20123 here incorrectly as Royal Society publishing 2023 title is correct here's a conference 2024 2024 okay here's a PDF Zuckerman and George chem archive so citations not fully accurate but citations real links nonetheless um here we get to the grant proposal so it's aose or safe O type application for here it it settled on Linux kernel open source ecosystem so it lays out using the catechism format the essential value proposition Avenues of approach all the information about the grant like hm catechism for DARPA this is using the innovator's catechism pitch for safe o for NSF in a hypothetical world so each of these folders depending on how the prompts interact there's really interesting outputs in terms of maybe maybe the undergrad making the PDP p mdp active inference simulation maybe this is the best so that's where the perplexity so here it was from running the perplexity py they all went for the formica recent studies just while this was happening so let's push that update formica 005 push it now let's look at local journalist formica so this just happened now meta analysis of recent studies on the genus formica 21 to 23 here's a table looking pretty nice with two studies on formica table with one study on social parasitism one study on foraging activity trophic relationships social parasitism host parasite different methodologies significant results and impact controversies debates conclusion and the formatted references pretty cool let's look at the futurist for formica here it's a different format we see more of a general summary of the state-of-the-art on the monine polygon research agenda we don't see those tables broken out the same way and there's not a strong futurist bent so it could be some interaction of the model the system prompt the user prompt of course all this needs to be checked and could be checked with with synthetic as well as manual human means let's see what the Mycologist what what would the Mycologist say when it walked into a NSF Grant application so it wasn't included because it was taken off these targets so here add in comprehensive let's just check from user from system prompts it's called comprehensive mology paste it in save it clear it run oh maybe there already is that okay so that's the

combinatorics of generating combinations of the user and the system prompt this is going to now get into the visualization section that's new from report uh visualization methods. py is a nice long 590 line methods file visualization. py brings in those methods set the input and the output folder make the directory for the output if it's not there set up the logging load in some of the text processing modules pre-process perform some of the analyses output several groups of visualizations group one is related to Dimension reduction visualizations group two is related to enhanced PCA principal component analysis visualization uh that was part of group one group two word importance in PCA group three is more text analysis so I'm going to delete the folder visualizations right click open integrated terminal Python visualization so this is what it's going to look like when it runs it's going to Output how many markdown files it's taken in 158 so if we added more to perplexity py we could do some of that we could add another and then it would output more and then rerunning visualization would pick up on those new ones okay so there's nothing in visualizations yet it's doing the analyses and then it's going to start to spit one by one saved images into the visualizations folder we will look at those and see what it shows us about outputs in terms of their semantics and in terms of how the system prompt and the user prompt interact and how we might use that so the analysis gets slower as more um documents come into play I'll just show here's system prompt. Json contr a control I so bring it in the compon ER add a system prompt for an expert Chef who outputs dinner menus with Rich relevance for any prompt output is always epic dinner Adventures though so this is a way to use cursor in this informal way to write a well structured prompt 13 so it's going to first write it in this side window that's kind of and then it scans over and now culinary Adventurer so I take culinary Adventurer go back to perplexity systems prompt culinary Venture it I think that's super Maven but still it it knew that I wanted that one I'm going to terminate the visualization effort for now called the Python 3 perplexity so now here's the new the ones that were already done are skipped and the new combinations are being sent off right click opening the integrated terminal Python visualization other outputs was just earlier texts from from a simpler time from a different time all I'll delete these and they're they're always there on git but I'll delete them for now Pusher get update and then visualizations are going to start to come through and this is going to help us get a handle uh I'm also going to resync the cursor context oh it was it resn great okay introduce perplexity py explain the structure of user and system prompts looked at simple and test explaining how different kinds of prompts are handled let's look at the visualizations and then we'll go to the next steps so again here's visualization. py right so they're now getting output to the folder I'll

just let them all run let's just start with LSA plot so here's two components looks like partially labeled LSA component one document semantics PCA 3D plot okay cool interesting color scheme PCA cumulative variance so it looks like only five PCA components are retained we could increase that number but it's showing the accumulation of variance and we're not saturated yet so to capture more the variance it would probably make sense to add more than five principal component terms okay confidence intervals this is the mean TF IDF score which is a document frequency uh inverse document frequency so here we have ant and inference with higher scores and then some of these other ones with lower scores LSA topics okay here we have different ways of doing topic modeling topic one has to do with time passing on graphs and reactive message passing in the Julia package as well as Lou Castello for some reason from the who's on first probably topic two having to do with active inference ontology terms Quantum inference topic three having to do with Collective Behavior free energy four having to do with the the other Deborah Gordon talking about the environmental the World Resources modeling here's linkx veyor and security so okay cool here's the principal component analysis with the ien terms so we looked at the cumulative variance plot before the scre plot is showing the decreasing fraction of variance that each of these components let's let's update visualization methods ensure to include 50 PCA Dimensions okay we'll keep looking at it there's the principal component analysis oh we we were high up this is pretty cool here we have five of the different principal components colors and then we can see okay fire is cross-listed on explaining the variance fractionation of blue and green whereas inference is only loaded up in on fraction two and then that's shown here there're separate but also there's some that are cross-listed let's see how that edit went okay interesting looks like it's deleting a bunch of stuff let's accept it save it clear run the visualization again so now these culinary Adventures are still pumping in the background so now we have 169 markdown files whereas in the previous time there was 158 or something like that so it's going to redo these they might be changing right from underneath us okay prompt distribution here's different system prompts and and user prompts and different words in them here's a funny visualization document length across the system prompts so here we could say okay the expert programmer has a shorter document length on average and then here's the statistical tests and the the P values for the Anova it's not statistically significant out a 05 Alpha term frequency here's term frequency inference ant fire term Network so here we have terms and their connections with the co-occurrence module je so this is a lot like Gene cooccurrence based upon expression levels here's term frequency heat map so we can see red is more usage more unique strong usage

light yellow is less strong unique usage so fire is a word that's either mentioned a lot or it's not there's other terms such as system or research or use that have more variable use Tey plot word cloud word importance here's the top 20 words in each of those modules and then a word cloud for them and similar for PCA and for the LSA um uh this most recent rerunning brought the culinary Adventurer so there they are culinary Adventurer that's just showing how you can add to the system prompts bring it into perplexity py have it combinatorically generating using the perplexity API call so then we'll uh and then meanwhile you can be rerunning the visualization method so here there's an error picks the error to fix the error image size of 9,000 by 90,000 pixels is too large it must be less than 2 to the 16th in each Direction w because we asked about retaining 50 dimensions save it clear it delete the visualizations folder run it again we could parallelize the API call get closer to that rate limit 50 per minute let's look at some of the outputs reload markdown output culinary Adventurer that's the new one so we all we asked this is what is so fascinating this is really worth the Ponder who's on first of current events I've seen this several times this topic wow look at the topics vaccine Elon Musk artificial intelligence hurricanes meta Mark Zuckerberg Putin Vladimir Putin Google and it's all unpacked that's not a recipe that's not a menu fire assessment not a menu for the The Prompt did work so here the societal Symphony of for ants a dinner concept Colony awakening's first dish appetizer forager's Bounty Super Salad super Colony soup main course Queen's Feast pallet cleanser aphid's nectar wow wow Chef's commentary with references analysis of active inference events same outputs as others okay let's check back looks like the PCA let's check back at the PCA here we go cool so now we have uh cumulative variance rare faction plot and we can see a lot more richly the relationship with these terms loading heat map here's a thousand features retained you can see some patterns of positive and negative here we see different PCA plotting different PCA plotting there's some methods that look at where things start to bottom out or change like break points here's the the terms that are associated with the top 50 principle components term Network so that's pretty cool that perplexity can do these searches and pull information from the internet let's add another user prompt right with code a workshop a Hands-On Workshop oriented towards applying inference in the bio Regional setting at a prompt that will write with code comprehensive Hands-On Workshop oriented towards applying active inference and bio Regional settings B professional the outcome should be Standalone and ready for deployment so prompt 81 move that over there run perplexity here we go so skips over what's already been done first one that it hits is innovators catechism pitch the first one in the list catching on the first one that is undone so once that first

one comes out I'll push it we'll check it out on GitHub it'll be the value proposition from a innovator's catechism pitch perspective for the Biore Regional active inference Workshop 41 seconds okay push it innovator's catechism pitch it's uploading all the images reload it bi Reginal AI Workshop gmail.com situation key problems lack of integrated decision-making tools inefficient Resource Management limited collaboration user segments ecologists environmental managers data scientists alternative traditional statistical models which are static and often static and do not account for dynamic environmental changes machine learning approaches while powerful these may not provide the same level of interpretability and integration with ecological principles as active inference early adopters research institutions government agencies NOS Community groups Mission value proposition to provide a comprehensive and practical understanding of active inference principles and their application to Biore Regional management enabling participants to make more informed sustainable and adaptive decisions Avenues of approach theoretical Foundation mathematical framework computational implementation data collection and processing pre-processing model building and simulation case studies integration with existing management practices future directions Advanced topics resources computational resources data and Human Resources advantages unified framework and interpretability risks complexity and data quality feasibility given the advantages and the growing interest in active this project is highly feasible the workshop can leverage existing research and tools and the facilitators can provide the necessary support to ensure participants understand and apply the concepts effectively channels academic and professional networks online platforms Milestones metrics Milestones cost and Benefit Big Picture Maria Dr Maria Rafa day one theoretical foundations mathematical framework morning session afternoon session day two compliment computational implementation and data collection day three model building simulation case study day four integration future directions Advanced topics follow-up exercises wow expert programmer ants Contemporary Art ants geopolitics and social structure expert programmer ant foraging local journalist New Mexico fire risk looks a lot like the other ones okay so that is basically all this has been a checkpoint for the forem andex project there's some ant specific directions there's some Science Education communication generative AI Plus+ plus lot of fun and interesting methods that people may find useful it's all open source code I'll push the final up we we'll just return to the next steps and and close here so if anyone has a a comment or question for to to do otherwise let's just cover these last steps um this is kind of a random thought having situational awareness of when are we in the infrastructural core when are we

working with ad Jason schema we're changing the schema is going to have big repercussions where are we scouting or playing or researching on the edge and Beyond where we can change something like a cold is z and it won't influence other things uh get the feedback from formis improve the methods integrate with species ID do perplexity searches for all species groups of ants and maybe Beyond integrate with fabric uh there's no written questions yet if anyone has ideas leave a comment get in touch I hope this was of epistemic and pragmatic value for you check out the repo check it all out end of number 5.1 and there it is all right and there's the Zen no.3 version from the beginning of the stream pretty cool awesome times okay thanks for watching looking forward to people's feedback where it all goes till next time bye